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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte(Deemed to be University)
Second Semester M.C.A. (CBCS) Degree Examinations

Summer Semester September -2023
22MCA203 -OPERATING SYSTEM WITH UNIX

Duration: 3 Hours

Max. Marks:100

Note: 1) Answer Five full questions choosing One full question from each unit**Unit – I****Marks BT* CO* PO***

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|--|----|----|---|---|
| 1. a) What is a system call? Illustrate the use of system calls in writing a program to read data from one file and copy them to another file. | 10 | L1 | 1 | 1 |
| b) Describe the differences among short-term, medium-term, and long-term scheduling | 6 | L2 | 1 | 1 |
| c) Briefly explain the features of any two special purpose systems | 4 | L2 | 1 | 1 |
| | | | | |
| 2. a) Mention the advantages of layered approach to operating System design with neat diagram. | 10 | L2 | 1 | 1 |
| b) Interpret virtual machines and list its benefits. | 6 | L3 | 1 | 1 |
| c) Describe the Interprocess communication in operating system | 4 | L2 | 1 | 1 |

Unit – II

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|--|---|----|---|---|
| 3. a) How are signals handled in multithreaded programs? | 6 | L1 | 2 | 1 |
| b) Name the different scheduling criteria's. | 4 | L2 | 2 | 1 |
| c) Assume you have the following jobs to execute with one processor, Consider the set of Processes, Priorities, CPU burst time as given in table below | | | | |

Process	Burst time	Priority
1	10	3
2	1	1
3	2	3
4	1	4
5	5	2

Draw a Gantt chart and calculate average waiting time and turnaround time of each process, using FCFS, SJF, Priority and RR scheduling algorithms.

(Time quantum = 1) $\begin{pmatrix} 1 \text{ high priority} \\ 0 \text{ low priority} \end{pmatrix}$

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|---|----|----|---|---|
| 4. a) What is critical section problem? How can it be solved? | 10 | L3 | 2 | 2 |
| b) Define readers-writers problem. | 6 | L2 | 2 | 1 |
| c) Consider the following snapshot of the system | 4 | L2 | 2 | 1 |

	Allocation	Max	Available
	ABCD	ABCD	ABCD
P0	0012	0012	1520
P1	1000	1750	
P2	1354	2356	
P3	0632	0652	
P4	0014	0656	

Answer the following questions using Bankers Algorithm

- i) What is the content of Need Matrix?
ii) Is the system in Safe state?

10	L4	2	2
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P.T.O

Unit – III

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|----|----|--|----|----|---|---|
| 5. | a) | Swapping increases the operating system overhead. Justify. | 5 | L2 | 3 | 1 |
| | b) | Explain the Paging process. | 8 | L2 | 3 | 2 |
| | c) | Describe the Hashed Page tables in Paging. | 7 | L2 | 3 | 1 |
| 6. | a) | Compare global and local replacement schemes of frame allocation. | 5 | L2 | 3 | 1 |
| | b) | Consider a reference string 0,1,7,2,3,2,7,1,0,3 with four-page frames. How many page faults occur using page replacement algorithms: (i) FIFO (ii) LRU? Also describe the drawbacks of these algorithms. | 10 | L3 | 3 | 1 |
| | c) | Compare external and internal fragmentation. How can compaction solve the problem of external fragmentation? Explain. | 5 | L3 | 3 | 2 |

Unit – IV

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|----|----|---|----|----|---|---|
| 7. | a) | Explain 'chmod' with example. | 10 | L2 | 4 | 1 |
| | b) | Differentiate between hard link and soft link. | 10 | L1 | 4 | 1 |
| 8. | a) | Explain the following commands with various options
i) mkdir ii) pwd iii) cd | 6 | L1 | 4 | 1 |
| | b) | Describe Absolute Pathnames and Relative Pathnames. | 10 | L1 | 4 | 1 |
| | c) | List the two special Files in Redirection. | 4 | L1 | 4 | 2 |

Unit – V

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|----|----|--|----|----|---|---|
| 9. | a) | How do you Run Jobs in Background? Explain with examples | 10 | L2 | 5 | 1 |
| | b) | Mention the process in Job control and killing of Process. | 10 | L2 | 5 | 1 |
| 10 | a) | Describe the logical Operators && and . | 10 | L2 | 5 | 1 |
| | b) | Explain the Case statement in shell programming. | 10 | L2 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
